Research Aims

The primary aim of this research is to explore the cybersecurity threats associated with the Internet of Medical Things (IoMT) and to evaluate the challenges these threats pose to the healthcare sector. The study seeks to:

1. **Identify and categorize** the most common and emerging cybersecurity threats targeting IoMT devices in clinical and remote care settings.
2. **Analyse the vulnerabilities** within IoMT systems and their potential impact on patient safety, data confidentiality, and healthcare service continuity.
3. **Assess the current cybersecurity practices and defense mechanisms** adopted by healthcare organizations to mitigate these risks.
4. **Propose effective strategies and frameworks** for enhancing the cybersecurity posture of IoMT environments.
5. **Raise awareness** among stakeholders—including healthcare providers, policymakers, and technology developers—about the critical importance of securing IoMT infrastructure.

Research Methodology and Source Selection

1. Databases: IEEE Xplore, ScienceDirect, PubMed, SpringerLink
2. Keywords: “IoMT cybersecurity,” “medical device security,” “AI in healthcare IoT”
3. Timeframe: 2018–2024
4. Inclusion of peer-reviewed, technical reports, and global policy papers